

- 3 Migration is when many individuals of a species move from one geographical region to another.

Monarch butterflies migrate from Canada and North America to Mexico each autumn.

The photograph shows a cluster of migrating Monarch butterflies on a branch of a tree.



(Source: © Patrick Stoll/Alamy Stock Photo)

- (a) (i) Migration is an example of which type of adaptation?

(1)

- ☐ A anatomical
- ☐ B behavioural
- ☐ C isolational
- ☐ D physiological

- (ii) The heterozygosity index for a cluster of 8 000 Monarch butterflies was calculated as 0.166.

Calculate the number of heterozygotes in this cluster.

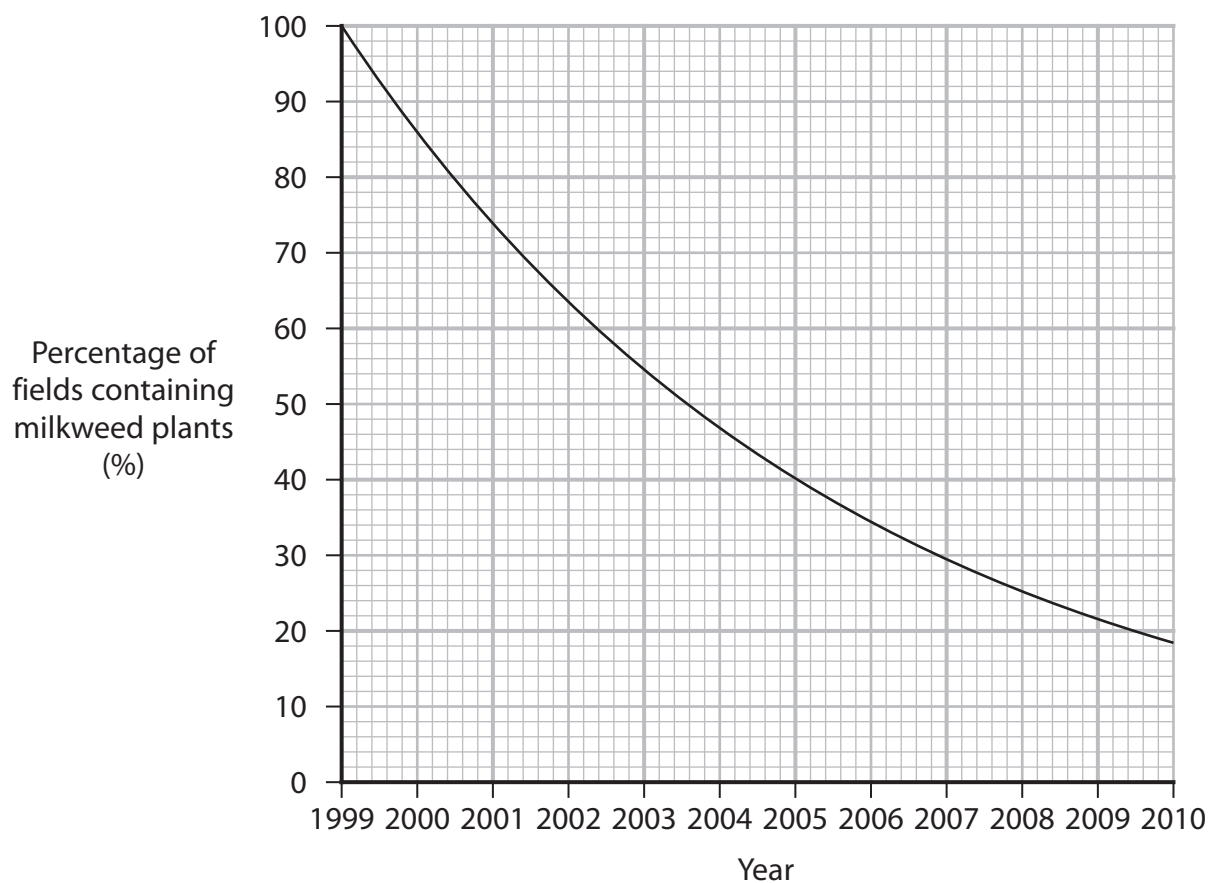
(1)

Answer

(b) Female Monarch butterflies lay fertilised eggs on the leaves of milkweed plants.

Herbicides can be used by farmers to kill plants such as milkweed. Farmers in the USA have increased the use of glyphosate herbicide since 1999 to kill plants, such as milkweed.

The graph shows the percentage of fields in the USA containing milkweed plants from 1999 to 2010.



Explain what effect this human activity would have had on populations of Monarch butterflies, including their genetic diversity, after 2010.

Use the information in the graph to support your answer.

(3)

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- (c) The photograph shows a hen (female chicken) that has just laid an egg.



(Source: © Angela Hampton Picture Library / Alamy Stock Photo)

The rate of egg laying by hens is an example of polygenic inheritance.

This phenotype shows continuous variation.

- (i) State what is meant by **polygenic inheritance** with reference to this phenotype.

(1)

- (ii) Sketch a **labelled** graph to show the expected egg laying rate for a population of hens.

(2)



(Total for Question 3 = 8 marks)

